**Nazorat testi** · *Aralash test · 10-variant* — *Milliy sertifikat 2026. Kimyo*

***Vaqt:*** *180 daqiqa*

*Davriy jadvaldan foydalanish mumkin.*

**1.** $CO + H_2O \rightleftarrows CO_2 + H_2$ reaksiyasi 4 l hajmli idishda o‘tkazildi. Ma’lumotlar jadvalda berilgan:

| Modda | CO | $H_2O$ | $CO_2$ |
|---|---|---|---|
| Boshlang‘ich, mol | 6 | 3 | 0 |
| Muvozanatda, mol | — | — | 2 |

Muvozanat **konstantasini (K) hisoblang.**

A) *2* B) *1* C) *0,5* D) *4*

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**2.** Rasmda neytral atom elektron qavatlarining orbital diagrammasi (kataklar – orbitallar, o‘qlar – elektronlar) berilgan.

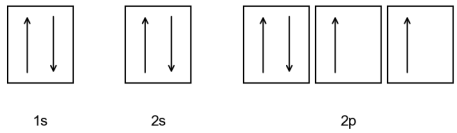



Diagramma qaysi element atomiga tegishli?

A) *ftor* B) *neon* C) *kislorod* D) *uglerod*

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**3.** Faqat oksidlovchi bo‘la oladigan **ionlarni aniqlang**: 1) $Zn^{2+}$; 2) $Fe^{2+}$; 3) $Cl^-$; 4) $Fe^{3+}$

A) *1, 2* B) *2, 4* C) *1* D) *1, 3*

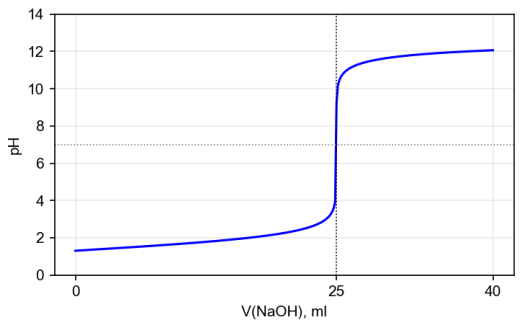
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**4.** Massa soni 226 va zaryadi 88 bo‘lgan yadro massa soni 206, zaryadi 82 bo‘lgan yadroga aylanishida nechta $\alpha$- va $\beta^-$-zarra chiqaradi?

A) *6α va 4β* B) *5α va 4β* C) *5α va 6β* D) *4α va 5β*

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**5.** 25 ml HCl eritmasi 0,05 mol/l li NaOH eritmasi bilan titrlandi. Titrlash egri chizig‘i grafikda berilgan (ekvivalent nuqta 25 ml da).




Grafikning boshlang‘ich nuqtasi ($V = 0$) pH ining qiymatiga mos keluvchi HCl eritmasidagi $H^+$ **konsentratsiyasini (mol/l) toping.**

A) *0,5* B) *0,005* C) *0,05* D) *0,1*

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**6.** 20 l NO va 30 l $O_2$ aralashmasi $2NO + O_2 \rightarrow 2NO_2$ tenglama bo‘yicha to‘liq reaksiyaga kirishdi (barcha gazlar bir xil sharoitda o‘lchandi). Aralashma hajmi necha litrga o‘zgargan (modul bo‘yicha)?

A) *10* B) *50* C) *40* D) *20*

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**7.** $2NO + O_2 \rightleftarrows 2NO_2$ reaksiyada muvozanat holatida $[NO] = 0,5$, $[O_2] = 2$, $[NO_2] = 4$ mol/l. Muvozanat **konstantasini (K) hisoblang.**

A) *64* B) *16* C) *4* D) *32*

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**8.** 11,2 g Fe va 9,6 g S aralashtirilib, reaksiya o‘tkazildi ($Fe + S \rightarrow FeS$). Hosil bo‘lgan FeS **massasini (g) aniqlang.**

A) *26,4* B) *35,2* C) *8,8* D) *17,6*

---

**9.** Quyidagi jadvalda $Na_2CO_3$ tuzining eritmalari keltirilgan:

| No. | m(eritma), g | ω, % |
|---|---|---|
| I | 150 | 12 |
| II | 100 | 30 |

Eritmalar aralashtirildi. Oxirgi eritmadagi tuzning massa **ulushini (%) toping.**

A) *21* B) *42* C) *22,8* D) *19,2*

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**10.** Tabiiy elementning ikki izotopi: $^{63}Cu$ (70%) va $^{65}Cu$ (30%). Elementning o‘rtacha nisbiy atom **massasini aniqlang.**

A) *64* B) *64,4* C) *63,6* D) *63*

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**11.** Quyidagi zarrachalardan qaysilari neon atomi bilan izoelektron? 1) $N^{3-}$; 2) $Al^{3+}$; 3) $Cl^-$; 4) $Ca^{2+}$

A) *2, 3, 4* B) *1, 2* C) *2, 3* D) *1, 3*

---

**12.** 2 mol $Ba(OH)_2$ va 2 mol $HNO_3$ suvda eritilib, eritma hajmi 2 l ga yetkazildi. Eritmadagi

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barcha ionlarning umumiy molyar **konsentratsiyasini (mol/l) aniqlang** (kuchli elektrolitlar to‘liq dissotsilanadi).
A) *2* B) *10* C) *2,5* D) *5*

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**13.** $Na_2SO_4 \cdot 10H_2O$ kristallogidratida kristallizatsion suvning massa **ulushini (%) aniqlang.**
A) *55,9* B) *44,1* C) *28* D) *126,8*

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**14.** Quyidagi tuzlardan qaysilari asosli tuz? 1) $K_2SO_4$; 2) $(CuOH)_2CO_3$; 3) $CaCO_3$; 4) $KAl(SO_4)_2$
A) *2, 4* B) *2* C) *2, 3* D) *4*

---

**15.** $2Al + 3CuCl_2 \rightarrow 2AlCl_3 + 3Cu$ reaksiyasida 6,75 g Al ishtirok etdi. Cu **miqdorini (mol) toping.**
A) *0,17* B) *0,75* C) *0,19* D) *0,38*

---

**16.** Nodir gazlar haqida qaysi fikrlar to‘g‘ri? 1) Geliy eng yengil inert gaz; 2) Neon reklama naylarida ishlatiladi; 3) Nodir gazlar odatiy sharoitda ko‘p atomli molekulalar hosil qiladi; 4) Argon havoning hajmi bo‘yicha eng ko‘p uchraydigan inert gazi
A) *1, 2* B) *1, 2, 4* C) *1, 2, 3* D) *3, 4*

---

**17.** 47 g $Cu(NO_3)_2$ to‘liq parchalanguncha qizdirildi. Qattiq qoldiq **massasini (g) aniqlang.**
A) *47* B) *40* C) *10* D) *20*

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**18.** 60 g $FeS_2$ yondirilib, hosil bo‘lgan $SO_2$ dan $H_2SO_4$ olindi (jami unum 80%). Sulfat kislota **massasini (g) aniqlang.**
A) *78,4* B) *49* C) *156,8* D) *98*

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**19.** $Zn(NO_3)_2$ qizdirilganda qanday moddalar hosil bo‘ladi?
A) *Ag, $NO_2$ va $O_2$* B) *ZnO, $NO_2$ va $O_2$*
C) *$NaNO_2$ va $O_2$* D) *$KNO_2$ va $O_2$*

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**20.** 25,25 g $CH_3Cl$ dan quyidagi ikki bosqichda HCl olinadi: 1) $2CH_3Cl + 2Na \rightarrow C_2H_6 + 2NaCl$ (unum 90%); 2) $C_2H_6 + Cl_2 \rightarrow C_2H_5Cl + HCl$ (unum 50%). Oxirgi mahsulot HCl ning **hajmini (l, n.sh.) toping.**
A) *2,8* B) *5,04* C) *5,6* D) *2,52*

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**21.** $CH_2{=}CH{-}CH_2{-}CH_3$ birikmaning sistematik nomi qaysi qatorda to‘g‘ri keltirilgan?
A) *buten-1* B) *buten-2* C) *1-butin* D) *butan*

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**22.** Arenlar, neft va tabiiy gaz haqida qaysi fikrlar to‘g‘ri? 1) Benzol bromli suvni rangsizlantirmaydi; 2) Tabiiy gazning asosiy qismi propan; 3) Neft fraksiyalaridan gazolin 150 °C gacha haydaladi; 4) Benzoldan nitrobenzol nitrolash bilan olinadi
A) *3, 4* B) *1, 2* C) *1, 3, 4* D) *2, 3, 4*

---

**23.** $CH_3CHO + Ag_2O\ (NH_3)$ reaksiyasi natijasida qaysi organik modda hosil bo‘ladi?
A) *etilatsetat* B) *etanol*
C) *sirka kislota* D) *metan*

---

**24.** Aldegid va ketonlar haqida qaysi fikrlar to‘g‘ri? 1) Formaldegid gaz holatda, suvda yaxshi eriydi; 2) Ketonlar kuchsiz oksidlovchilar bilan oksidlanmaydi; 3) Aldegidlar qaytarilganda ikkilamchi spirt hosil bo‘ladi; 4) Sirka aldegid Kucherov reaksiyasi bilan olinadi
A) *1, 2, 4* B) *1, 4* C) *1, 2* D) *2, 3*

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**25.** $2HCOOH + Mg \rightarrow (HCOO)_2Mg + H_2$ jarayoni uchun 18,4 g HCOOH olingan. $H_2$ **hajmini (l, n.sh.) aniqlang.**
A) *4,48* B) *17,92* C) *8,96* D) *2,24*

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**26.** Murakkab efirlar va yog‘lar haqida qaysi fikrlar to‘g‘ri? 1) Efirlanish reaksiyasi qaytmas; 2) Etilatsetat efir hidli suyuqlik; 3) Suyuq yog‘larni gidrogenlab qattiq yog‘ olinadi; 4) Suyuq yog‘lar to‘yinmagan kislotalardan hosil bo‘lgan
A) *2, 3, 4* B) *1, 2* C) *1, 4* D) *2, 4*

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**27.** Reaksiya tenglamasi: $C_{12}H_{22}O_{11} + H_2O \rightarrow 2C_6H_{12}O_6$. Reaksiyaga 684 g $C_{12}H_{22}O_{11}$ kirishdi. $C_6H_{12}O_6$ **massasini (g) hisoblang.**
A) *180* B) *720* C) *1440* D) *360*

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**28.** 26,7 g alanin HCl bilan reaksiyaga kirishganda hosil bo‘lgan tuz **massasini (g) aniqlang.**
A) *26,7* B) *37,65* C) *10,95* D) *75,3*

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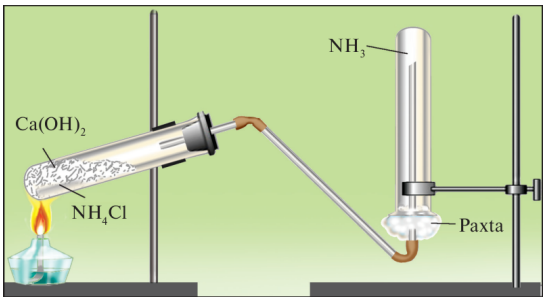
**29.** $C_2H_4 + Cl_2 \rightarrow C_2H_4Cl_2$ reaksiyasida 4,48 l (n.sh.) $Cl_2$ ishtirok etdi. $C_2H_4Cl_2$ **massasini (g) toping.**
A) *19,8* B) *39,6* C) *9,9* D) *29,7*

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**30.** Eritmaga NaOH (qizdirilganda) qo‘shilganda o‘tkir hidli gaz ($NH_3$) kuzatildi. Eritmada qaysi ion bor?
A) $Ba^{2+}$ B) $SO_4^{2-}$ C) $NH_4^+$ D) $I^-$

**31.** Metiloranj indikatori ishqoriy muhitda qanday rang beradi? A) *ko‘k*
B) *malina rangga kiradi (pushti)*
C) *rangsiz*
D) *sariq*

**32.** Rasmdagi asbobda olingan ammiak qanday yig‘iladi?




A) *og‘zi yuqoriga qaratilgan idishda*
B) *suvni siqib chiqarish usulida*
C) *og‘zi pastga qaratilgan probirkada, havoni siqib chiqarish usulida*
D) *yig‘ib bo‘lmaydi* —

| | |
|---|---|
| **Topshiriqlar (33–35) va javob variant (A–F)larini o‘zaro to‘g‘ri moslashtiring.** Sirka kislota va etanoldan iborat 16,6 g aralashmaga mo‘l $NaHCO_3$ eritmasi qo‘shilganda 4,48 l (n.sh.) gaz ajraldi. Xuddi shunday massadagi aralashma natriy bilan ta’sirlashganda esa 3,36 l (n.sh.) vodorod ajraldi. | A *9,6*<br>B *6,9*<br>C *12*<br>D *4,6*<br>E *26,24*<br>F *13,12* |
| **33. Aralashmadagi sirka kislota massasini (g) aniqlang.** | |
| **34. Aralashmadagi etanol massasini (g) aniqlang.** | |
| **35. Aralashma to‘liq neytrallanganda (faqat kislota) hosil bo‘lgan natriy atsetat massasini (g) aniqlang (reaksiya unumi 80%).** | |

**36.** $S + HNO_3 \rightarrow H_2SO_4 + NO$ reaksiya tenglamasini tenglang va barcha koeffitsiyentlar **yig‘indisini aniqlang.**

**Javob:** _____

**Diqqat!** Javobingizni javoblar varaqasiga ko‘chirib yozing.

**37.** $CH_4$ va $H_2$ dan iborat 60 ml aralashma mo‘l kisloroda yondirildi (suv suyuq holatda; hajmlar bir xil sharoitda o‘lchandi). Reaksiyadan so‘ng gazlar hajmi 100 ml ga kamaydi. Aralashmadagi $CH_4$ **hajmini (ml) toping.**

**Javob:** _____

**Diqqat!** Javobingizni javoblar varaqasiga ko‘chirib yozing.

**38.** 29,6 g $Mg(NO_3)_2$ to‘liq parchalanguncha qizdirildi. Qattiq qoldiq **massasini (g) aniqlang.**

**Javob:** _____

**Diqqat!** Javobingizni javoblar varaqasiga ko‘chirib yozing.

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**39.** To‘yingan bir atomli karbon kislotaning 4,6 g massasi mo‘l $NaHCO_3$ eritmasi bilan reaksiyaga kirishganda 2,24 l (n.sh.) $CO_2$ ajraldi. Kislotaning molyar **massasini (g/mol) aniqlang.**

**Javob:** _____

**Diqqat!** Javobingizni javoblar varaqasiga ko‘chirib yozing.

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**40.** 5,2 g asetilendan Kucherov reaksiyasi bilan (80% unum) sirka aldegid olindi. Aldegid **massasini (g) aniqlang.**

**Javob:** _____

**Diqqat!** Javobingizni javoblar varaqasiga ko‘chirib yozing.

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## YOZMA ISH

**41. Topshiriqni bajarish jarayonida barcha reaksiya tenglamalarini yozing. Miqdoriy va arifmetik hisoblashlarni (unga asos bo‘lgan usul yoki formulani ko‘rsatgan holda) uzviy ketma-ketlikda bajaring.**

HCOONa tuzining 0,01 mol/l li suvli eritmasi berilgan. HCOOH ning dissotsilanish konstantasi Ka = $1{\cdot}10^{-}4$, K(suv) = $1{\cdot}10^{-14}$.

1)Tuzning gidrolizlanish reaksiyasi tenglamasini yozing va gidroliz konstantasini (Kh) hisoblang.

2)Eritmadagi $OH^-$ ionlarining konsentratsiyasini (mol/l) toping. pOH ni ham hisoblang.

3)Eritmaning pH ini hisoblang.

4)Gidrolizlanish darajasini (%) aniqlang. Eritmadagi $H^+$ konsentratsiyasini (mol/l) ham toping.

5)Shunday eritmadan 500 ml tayyorlash uchun necha gramm HCOONa kerak? Eritma 10 marta suyultirilsa gidrolizlanish darajasi necha marta ortadi?

**Diqqat!** Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

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**42.** Quyidagi o‘zgarishlar zanjirida $X$, $Y$, $Z$ noma’lum organik moddalar:

$$CH_4 \xrightarrow{Cl_2,\ \text{yorug‘lik}} X \xrightarrow{Na} Y \xrightarrow{Cl_2,\ \text{yorug‘lik}} Z \xrightarrow{KOH\ (\text{spirtli})} C_2H_4$$

1)$CH_4 \to X$ : reaksiya tenglamasini yozing; hosil bo‘lgan moddani nomlang.

2)$X \to Y$ : reaksiya tenglamasini yozing; hosil bo‘lgan moddani nomlang.

3)$Y \to Z$ : reaksiya tenglamasini yozing; hosil bo‘lgan moddani nomlang.

4)$Z \to C_2H_4$ : reaksiya tenglamasini yozing; hosil bo‘lgan moddani nomlang.

**Diqqat!** Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

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**43. Topshiriqni bajarish jarayonida barcha reaksiya tenglamalarini yozing. Miqdoriy va arifmetik hisoblashlarni (unga asos bo‘lgan usul yoki formulani ko‘rsatgan holda) uzviy ketma-ketlikda bajaring.**

Uch probirkada $XCl_2$, $YCl_2$ va $ZCl_2$ tarkibli metall xloridlarining eritmalari bor. Har bir eritmaga reagentlar ta’sir ettirildi, kuzatuvlar jadvalda ("—" belgisi reaksiya bormasligini bildiradi, bo‘sh kataklar bo‘yicha ma’lumot berilmagan):

| **Reagent** | **$XCl_2$** | **$YCl_2$** | **$ZCl_2$** |
|---|---|---|---|
| $H_2S$ | qora cho‘kma | — | |
| $Na_2S$ | qora cho‘kma | qora cho‘kma | oq cho‘kma |
| NaOH | havorang cho‘kma | och yashil cho‘kma | oq cho‘kma |
| Zn plastinka | qizil qoplama | | — |

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$X$, $Y$, $Z$ — davriy sistemaning to‘rtinchi davridagi metallar.
1)Jadvalda kuzatilgan barcha reaksiyalarning molekulyar tenglamalarini yozing. $XCl_2$ eritmasiga tushirilgan Zn plastinkada 0,1 mol $X$ ajralgan bo‘lsa, plastinka massasi qanchaga (g) o‘zgargan?
2)$X$, $Y$, $Z$ metallarni aniqlang va tuzlarning formulalarini yozing. $XCl_2$ tarkibida $X$ ning massa ulushi 47,4% ekanini hisob bilan tasdiqlang.
3)12,7 g $YCl_2$ eritmasiga mo‘l NaOH qo‘shildi, hosil bo‘lgan cho‘kma havoda to‘liq oksidlanguncha qoldirildi. Barcha reaksiya tenglamalarini yozing va oxirgi cho‘kma massasini (g) aniqlang.
4)$ZCl_2$ ning 13,6 g massasi mo‘l NaOH eritmasi bilan reaksiyaga kirishganda hosil bo‘lgan kompleks tuz massasini (g) hisoblang (reaksiya tenglamalarini yozing).
5)Hosil bo‘lgan $Cu(OH)_2$ cho‘kmasi (0,1 mol) qizdirilganda hosil bo‘ladigan CuO massasini (g) hisoblang.

**Diqqat!** Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

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## Nazorat testi · *10-variant* **Javoblar kaliti**

*Faqat o‘qituvchi uchun.*

**1–32-savollar**

| **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | **12** | **13** | **14** | **15** | **16** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *B* | *C* | *C* | *B* | *C* | *A* | *D* | *D* | *D* | *C* | *B* | *D* | *A* | *B* | *D* | *B* |
| **17** | **18** | **19** | **20** | **21** | **22** | **23** | **24** | **25** | **26** | **27** | **28** | **29** | **30** | **31** | **32** |
| *D* | *A* | *B* | *D* | *A* | *C* | *C* | *A* | *A* | *A* | *B* | *B* | *A* | *C* | *D* | *C* |

**33–35-savollar**

| No | Javob | Qiymat |
|---|---|---|
| 33 | *C* | 12 |
| 34 | *D* | 4,6 |
| 35 | *F* | 13,12 |

**36–40-savollar**

| No | To‘g‘ri javob |
|---|---|
| 36 | 6 |
| 37 | 20 |
| 38 | 8 |
| 39 | 46 |
| 40 | 7,04 |

**41–43-topshiriqlar: javoblar**

**41-topshiriq.**

1) $1 \cdot 10^{-}10$
2) $1 \cdot 10^{-}6$ mol/l; 6
3) pH = 8
4) 0,01%; 1,00e-08 mol/l
5) 0,34 g; 3,16 marta

**42-topshiriq.**

1) $CH_4 + Cl_2$, yorug‘lik $\rightarrow CH_3Cl$ (xlormetan)
2) $CH_3Cl + Na \rightarrow C_2H_6$ (etan)
3) $C_2H_6 + Cl_2$, yorug‘lik $\rightarrow C_2H_5Cl$ (xloretan)
4) $C_2H_5Cl$ + KOH (spirtli) $\rightarrow C_2H_4$ (etilen)

**43-topshiriq.**

1) −0,1 g
2) 47,4%
3) 10,7 g
4) 17,9 g
5) 8 g

### Mavzular taqsimoti

| No | Mavzu | Tekshiriladigan ko‘nikma | Ball |
|---|---|---|---|
| 1 | Tirkashev VI bob (muvozanat, jadval) | Jadval tahlili | 1 |
| 2 | 8-sinf (elektron konfiguratsiya, orbital diagramma) | Rasm bo‘yicha tahlil | 1 |
| 3 | Tirkashev *V* bob; 11-sinf 7-bob | Bilim va tushunish | 1 |
| 4 | 11-sinf 1-bob (yadro reaksiyalari) | Bilim va tushunish | 1 |
| 5 | Tirkashev *V* bob (pH, grafik) | Grafik tahlili | 1 |
| 6 | 8-sinf (Gey-Lyussak qonuni) | Hisoblash va masala yechish | 1 |
| 7 | Tirkashev VI bob (kimyoviy muvozanat) | Hisoblash va masala yechish | 1 |
| 8 | 8-sinf (ortiqcha modda masalasi) | Hisoblash va masala yechish | 1 |
| 9 | Tirkashev III bob (eritmalar, jadval) | Jadval tahlili | 1 |
| 10 | 8-sinf (izotoplar) | Hisoblash va masala yechish | 1 |

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| | | | |
|---|---|---|---|
| 11 | Tirkashev I bob; 8-sinf (izoelektron) | Bilim va tushunish | 1 |
| 12 | Tirkashev IV bob (kuchli elektrolitlar) | Hisoblash va masala yechish | 1 |
| 13 | 7-sinf; Tirkashev (kristallogidrat) | Hisoblash va masala yechish | 1 |
| 14 | 8-sinf; 9-sinf (tuzlar sinflari) | Bilim va tushunish | 1 |
| 15 | 9-sinf; Tirkashev (genetik zanjir/reaksiya turlari, hisob) | Hisoblash va masala yechish | 1 |
| 16 | Tashev (nodir gazlar); 8-sinf | Bilim va tushunish | 1 |
| 17 | Tashev (nitratlar termik parchalanishi) | Hisoblash va masala yechish | 1 |
| 18 | Tashev ($H_2SO_4$ olinishi) | Hisoblash va masala yechish | 1 |
| 19 | Tashev (nitratlar) | Bilim va tushunish | 1 |
| 20 | 10-sinf (ikki bosqichli hisob) | Hisoblash va masala yechish | 1 |
| 21 | 10-sinf (nomenklatura §6, §11, §20, §24–26) | Bilim va tushunish | 1 |
| 22 | 10-sinf (mavzuga oid paragraflar); Abdusamatov | Bilim va tushunish | 1 |
| 23 | 10-sinf (olinish usullari) | Bilim va tushunish | 1 |
| 24 | 10-sinf (mavzuga oid paragraflar); Abdusamatov | Bilim va tushunish | 1 |
| 25 | 10-sinf (kis, hisob) | Hisoblash va masala yechish | 1 |
| 26 | 10-sinf (mavzuga oid paragraflar); Abdusamatov | Bilim va tushunish | 1 |
| 27 | 10-sinf (ugl, hisob) | Hisoblash va masala yechish | 1 |
| 28 | 10-sinf §34 | Hisoblash va masala yechish | 1 |
| 29 | 10-sinf (pol, hisob) | Hisoblash va masala yechish | 1 |
| 30 | 9-sinf (sifat reaksiyalar) | Bilim va tushunish | 1 |
| 31 | 8-sinf (indikatorlar) | Bilim va tushunish | 1 |
| 32 | 8-sinf (23-rasm) | Rasm bo‘yicha tahlil | 1 |
| 33–35 | Aralash masala (moslashtirish) | Ko‘p bosqichli hisob | 3 |
| 36–40 | Qisqa javobli masalalar | Hisoblash | 5 |
| 41–43 | Yozma ish | Usul va hisob (M, A) | 75 |

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